

Problem Set 2 - IO

Tate Mason

Question 1

Exploring the Data

```
# Load necessary libraries
library(dplyr)
library(gtsummary)
library(ggplot2)
library(readr)
library(lubridate)
library(here)
library(kableExtra)
library(modelsummary)
library(gmm)
library(tidyr)
# Load the dataset
data <- read_csv(here("Y2S1/IO/ps2/data", "prod_data_all.csv"))
# Clean dataset to harmonize year discrepancy - 1981 = 81
data <- data %>%
  mutate(year = ifelse(year < 100, year + 1900, year)) %>%
  filter(Y2 > 0, L > 0, K > 0, M > 0)%>%
  drop_na()

# Display the first few rows of the dataset
head(data)
```

```
# A tibble: 6 x 9
   id year      Y    Y2      L      K      M industry country
<dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <chr>    <chr>
1 10250 1981  296.  113.  2.86  15.5  183. metal    col
2 10732 1989 4733. 2566. 86.2  470. 2167. metal    col
```

3	10732	1990	4800.	2608.	73.6	644.	2192.	metal	col
4	10962	1982	7090.	1836.	37.9	5538.	5254.	metal	col
5	10962	1983	4942.	1343.	40.5	6314.	3599.	metal	col
6	11192	1981	1813.	1311.	24.0	1723.	502.	metal	col

1.1

```
# Summary statistics for each country - display no. obs for each sector and each sector over

data_col <- data %>% filter(country == "col")
data_chi <- data %>% filter(country == "chile")

# Number of observations by sector for each country
obs_by_sector_col <- data_col %>%
  group_by(industry) %>%
  summarise(n_obs = n()) %>%
  arrange(desc(n_obs))
obs_by_sector_chi <- data_chi %>%
  group_by(industry) %>%
  summarise(n_obs = n()) %>%
  arrange(desc(n_obs))
print("Colombia Observations by Sector:")
```

```
[1] "Colombia Observations by Sector:"
```

```
print(obs_by_sector_col)
```

```
# A tibble: 5 x 2
  industry n_obs
  <chr>    <int>
1 food      6140
2 apparel   4454
3 metal     3678
4 text      2847
5 wood       964
```

```
print("Chile Observations by Sector:")
```

```
[1] "Chile Observations by Sector:"
```

```
print(obs_by_sector_chi)
```

```
# A tibble: 5 x 2
  industry n_obs
  <chr>    <int>
1 food      18833
2 metal      5856
3 text       5447
4 wood       4835
5 apparel    4694
```

```
# Number of observations by year for each sector in each country
obs_by_year_sector_col <- data_col %>%
  group_by(year, industry) %>%
  summarise(n_obs = n()) %>%
  arrange(year, industry)
obs_by_year_sector_chi <- data_chi %>%
  group_by(year, industry) %>%
  summarise(n_obs = n()) %>%
  arrange(year, industry)
print("Colombia Observations by Year and Sector:")
```

```
[1] "Colombia Observations by Year and Sector:"
```

```
print(obs_by_year_sector_col)
```

```
# A tibble: 55 x 3
# Groups:   year [11]
   year industry n_obs
  <dbl> <chr>    <int>
1  1981 apparel     694
2  1981 food       872
3  1981 metal      558
4  1981 text       429
5  1981 wood       160
6  1982 apparel    619
7  1982 food      803
8  1982 metal      514
9  1982 text      364
10 1982 wood      146
# i 45 more rows
```

```
print("Chile Observations by Year and Sector:")
```

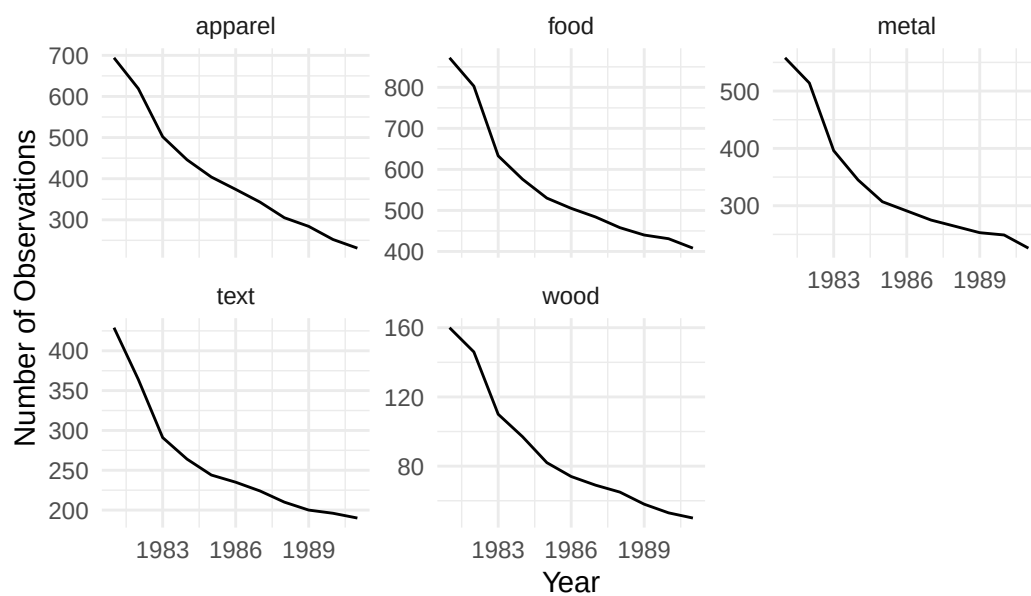
```
[1] "Chile Observations by Year and Sector:"
```

```
print(obs_by_year_sector_chi)
```

```
# A tibble: 90 x 3
# Groups:   year [18]
   year industry n_obs
  <dbl> <chr>    <int>
1  1979 apparel    354
2  1979 food     1171
3  1979 metal     373
4  1979 text      391
5  1979 wood      340
6  1980 apparel    317
7  1980 food     1067
8  1980 metal     350
9  1980 text      356
10 1980 wood      306
# i 80 more rows
```

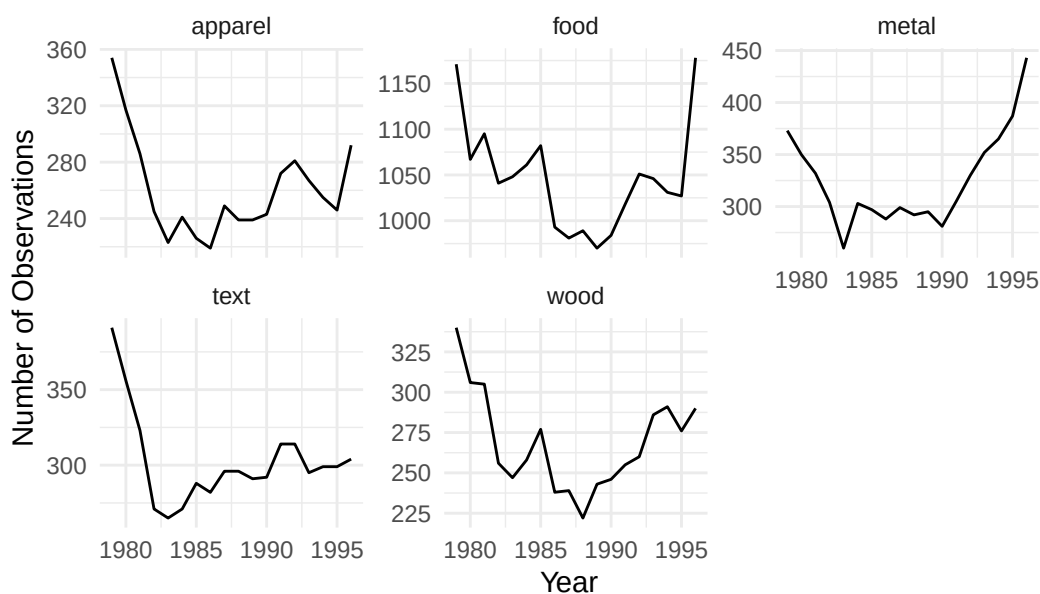
```
ggplot(data_col, aes(x = year, y = ..count..)) +
  geom_line(stat = "count") +
  facet_wrap(~ industry, scales = "free_y") +
  labs(title = "Colombia: Observations by Year and Industry",
       x = "Year",
       y = "Number of Observations") +
  theme_minimal()
```

Colombia: Observations by Year and Industry



```
ggsave("1.1_col.pdf", width = 10, height = 6)
ggplot(data_chi, aes(x = year, y = ..count..)) +
  geom_line(stat = "count") +
  facet_wrap(~ industry, scales = "free_y") +
  labs(title = "Chile: Observations by Year and Industry",
       x = "Year",
       y = "Number of Observations") +
  theme_minimal()
```

Chile: Observations by Year and Industry



```
ggsave("1.1_chi.pdf", width = 10, height = 6)
```

Chile: 1. Food - 2584, 2. Metal - 1038, 3. Wood - 867, 4. Apparel - 856, 5. Textile - 836

Colombia: 1. Food - 908, 2. Apparel - 744, 3. Metal - 632, 4. Textile - 475, 5. Wood - 173

Apparel Over Time - 1979: 354 → 1996: 292 - Peak at 1981 - 979

Food Over Time - 1979: 1171 → 1996: 1178 - Peak at 1981 - 1951

Metal Over Time - 1979: 373 → 1996: 443 - Peak at 1981 - 890

Textiles Over Time - 1979: 391 → 1996: 304 - Peak at 1981 - 752

Wood Over Time - 1979: 340 → 1996: 290 - Peak at 1981 - 465

1.2

```
# Creation of Investment Variable with 10 per depreciation rate
data <- data %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    dep      = 0.10,
    k_next = lead(K),
```

```

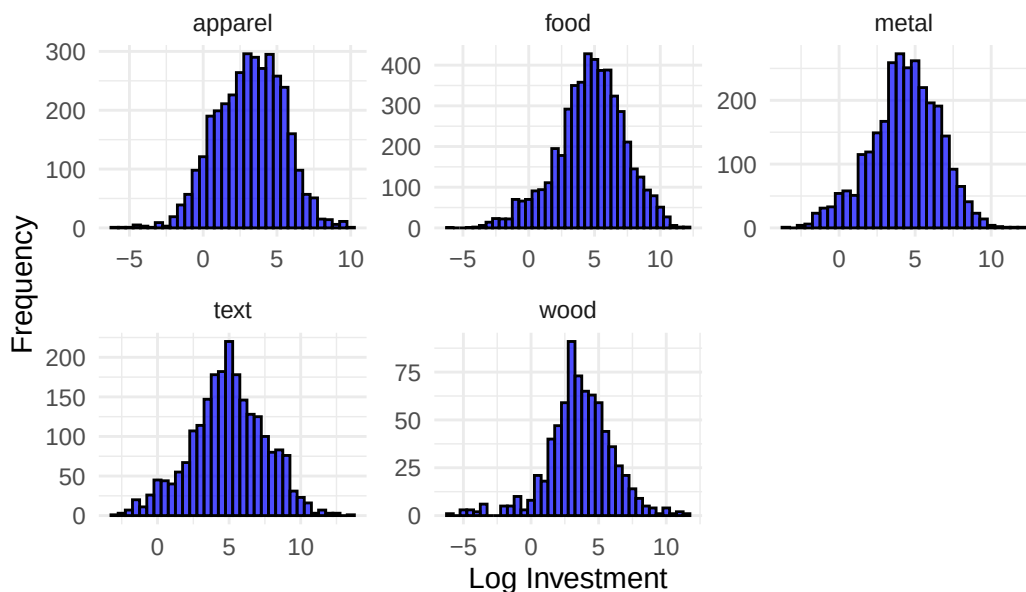
    i_raw = k_next - (1 - dep) * K,
    log_i = ifelse(i_raw > 0, log(i_raw), NA_real_),
    log_L = ifelse(L > 0, log(L), NA_real_),
    log_K = ifelse(K > 0, log(K), NA_real_),
    log_Y2 = ifelse(Y2 > 0, log(Y2), NA_real_)
  ) %>%
  ungroup()

data_col <- data %>% filter(country == "col")
data_chi <- data %>% filter(country == "chile")

# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Investment by Industry, Colombia",
       x = "Log Investment",
       y = "Frequency") +
  theme_minimal()

```

Histogram of Log Investment by Industry, Colombia



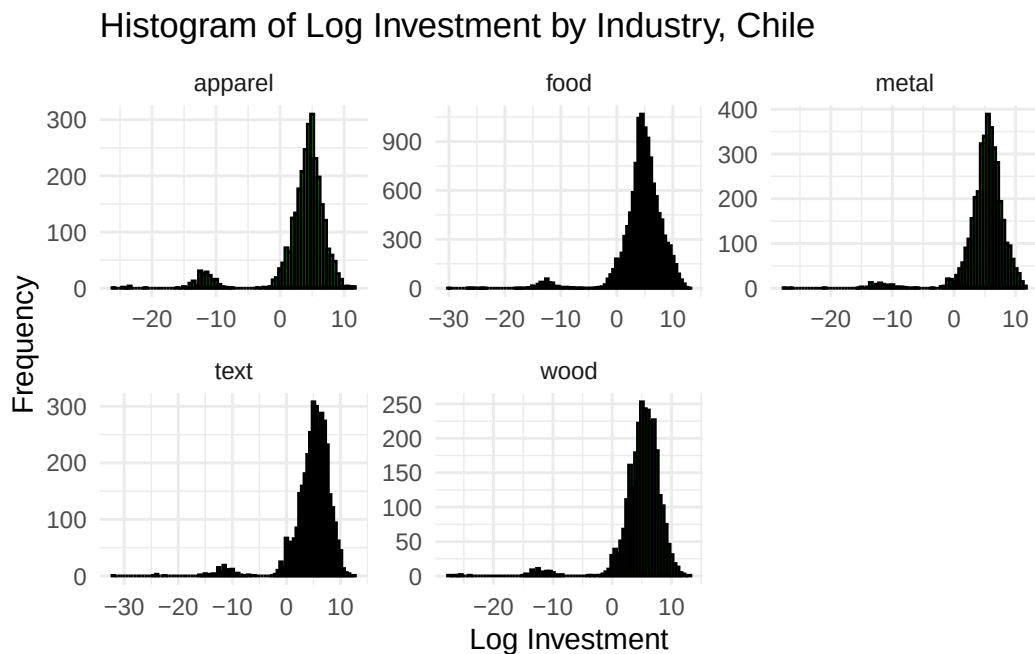
```

ggsave("1.2_col.pdf", width = 10, height = 6)

ggplot(data_chi, aes(x = log_i)) +

```

```
geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
facet_wrap(~ industry, scales = "free") +
labs(title = "Histogram of Log Investment by Industry, Chile",
      x = "Log Investment",
      y = "Frequency") +
theme_minimal()
```



```
ggsave("1.2_chi.pdf", width = 10, height = 6)
```

1.3

```
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()

data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
```



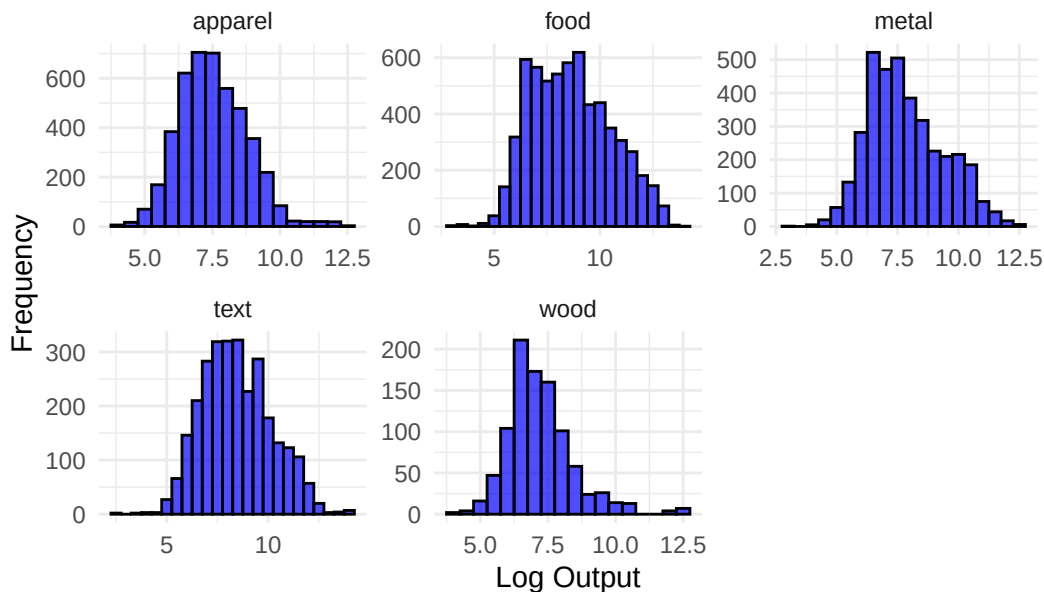
```

    log_Y = log(Y),
  ) %>%
  ungroup()

# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry, Colombia",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()

```

Histogram of Log Output by Industry, Colombia



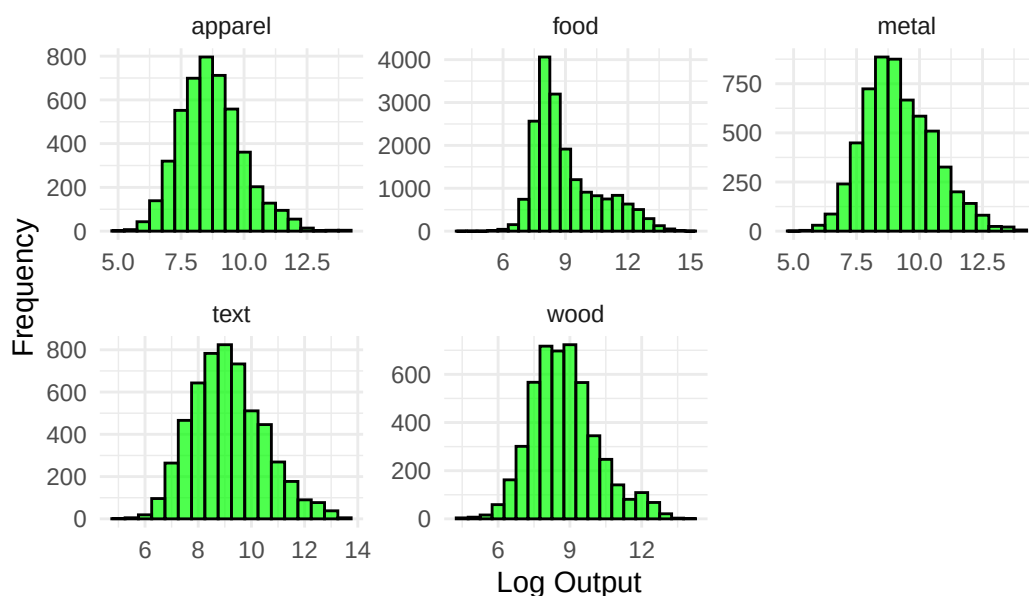
```

ggsave("1.3_col.pdf", width = 10, height = 6)

ggplot(data_chi, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry, Chile",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()

```

Histogram of Log Output by Industry, Chile

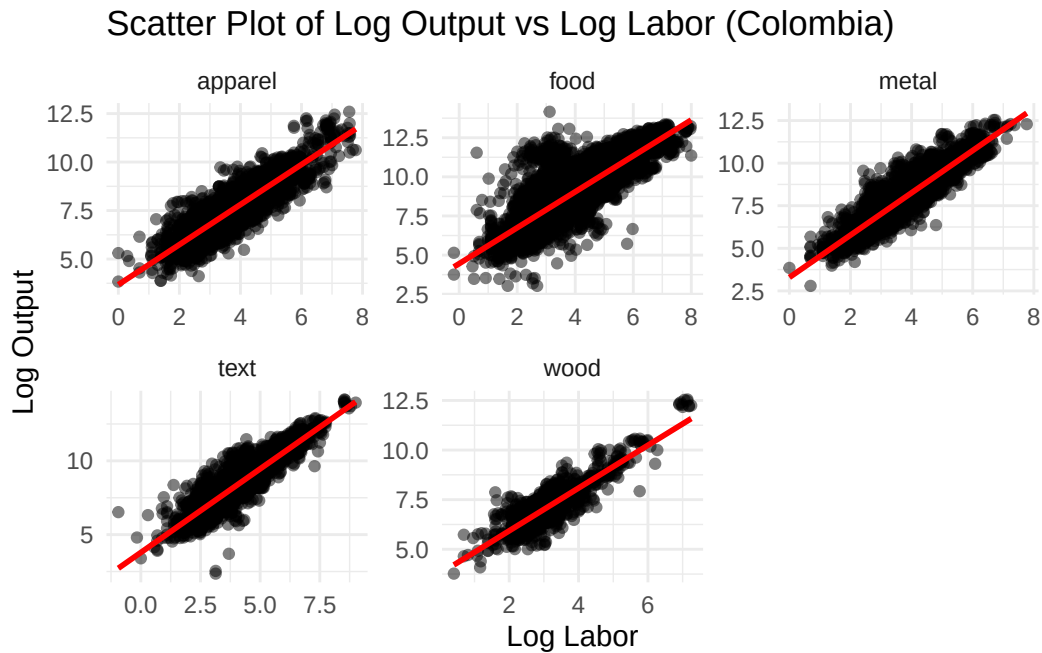


```
ggsave("1.3_chi.pdf", width = 10, height = 6)
```

1.4

```
# Scatter plot of log_Y and log_L for each sector in each country
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_col = log(K),
    log_L_ind_col = log(L)
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_chi = log(K),
    log_L_ind_chi = log(L)
  ) %>%
  ungroup()
```

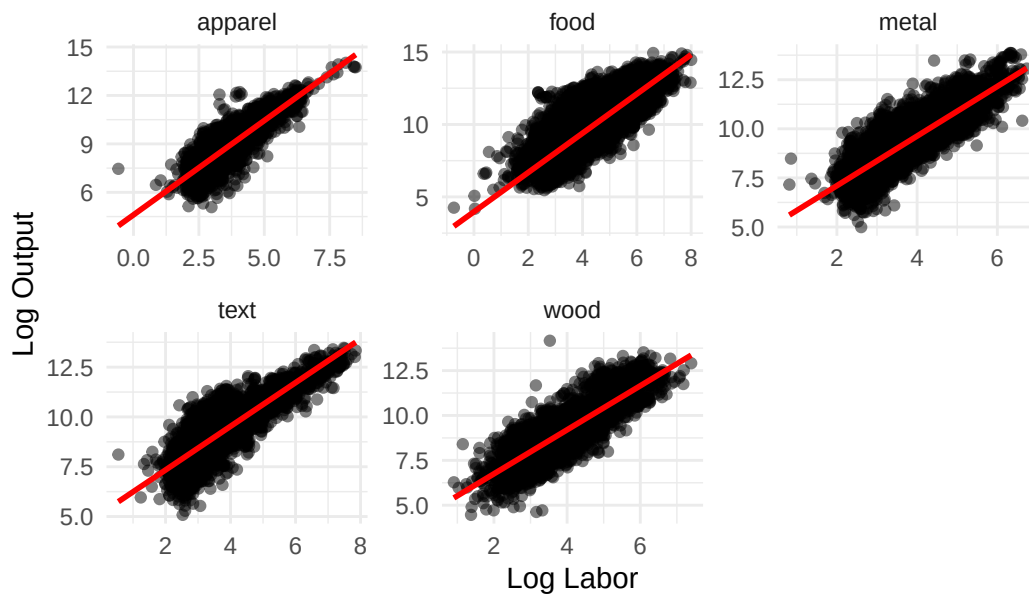
```
ggplot(data_col, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Colombia)",
       x = "Log Labor",
       y = "Log Output") +
  theme_minimal()
```



```
ggsave("1.4_col.pdf", width = 10, height = 6)

ggplot(data_chi, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Chile)",
       x = "Log Labor",
       y = "Log Output") +
  theme_minimal()
```

Scatter Plot of Log Output vs Log Labor (Chile)



```
ggsave("1.4_chi.pdf", width = 10, height = 6)
```

Question 2

2.1

```
# Estimate homogenous parameters using OLS for each country
ols_col <- lm(log(Y2) ~ log_L + log_K, data = data_col)

ols_chi <- lm(log(Y2) ~ log_L + log_K, data = data_chi)

modelsummary(
  list(Colombia = ols_col, Chile = ols_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

```
# Estimate varying parameters across industries using OLS for each country

ols_ind_col <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_col)
```

	Colombia	Chile
(Intercept)	2.186*** (0.018)	2.351*** (0.020)
log_L	0.831*** (0.006)	0.887*** (0.007)
log_K	0.273*** (0.004)	0.303*** (0.004)
Num.Obs.	18 083	39 665
R2	0.823	0.660
R2 Adj.	0.823	0.660
AIC	291 626.5	725 786.1
BIC	291 657.7	725 820.5
Log.Lik.	-18 545.704	-54 890.486
F	42 113.270	38 426.063
RMSE	0.67	0.97

* p < 0.1, ** p < 0.05, *** p < 0.01

```
ols_ind_chi <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_chi)

modelsummary(
  list(Colombia_Ind = ols_ind_col, Chile_Ind = ols_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

2.2

```
form_hom <- as.formula("log(Y2) ~ log_L + log(K) + log_i")

ols_inv_col <- lm(form_hom, data = data_col)
ols_inv_chi <- lm(form_hom, data = data_chi)

modelsummary(
  list(Colombia = ols_inv_col, Chile = ols_inv_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

	Colombia_Ind	Chile_Ind
(Intercept)	2.510*** (0.040)	2.865*** (0.063)
industryfood	−0.272*** (0.049)	−0.915*** (0.068)
industrymetal	−0.536*** (0.058)	0.131 (0.082)
industrytext	−0.294*** (0.059)	0.156* (0.083)
industrywood	0.049 (0.089)	−0.377*** (0.085)
industryapparel × log_L	0.921*** (0.013)	1.055*** (0.021)
industryfood × log_L	0.754*** (0.010)	0.896*** (0.010)
industrymetal × log_L	0.998*** (0.016)	0.994*** (0.018)
industrytext × log_L	0.858*** (0.015)	0.874*** (0.017)
industrywood × log_L	0.974*** (0.030)	1.016*** (0.020)
industryapparel × log_K	0.133*** (0.009)	0.150*** (0.013)
industryfood × log_K	0.321*** (0.007)	0.334*** (0.005)
industrymetal × log_K	0.208*** (0.010)	0.232*** (0.009)
industrytext × log_K	0.257*** (0.010)	0.243*** (0.010)
industrywood × log_K	0.117*** (0.018)	0.174*** (0.010)
Num.Obs.	18 083	39 665
R2	0.831	0.689
R2 Adj.	0.831	0.689
AIC	290 826.2	722 221.0
BIC	290 951.0	722 358.4
Log.Lik.	−18 133.547	−53 095.913
F	6352.788	6275.729
RMSE	0.66	0.92

* p < 0.1, ** p < 0.05, *** p < 0.01

	Colombia	Chile
(Intercept)	2.282*** (0.023)	2.521*** (0.026)
log_L	0.828*** (0.007)	0.844*** (0.009)
log(K)	0.242*** (0.006)	0.299*** (0.005)
log_i	0.031*** (0.004)	0.017*** (0.002)
Num.Obs.	14 315	25 125
R2	0.836	0.689
R2 Adj.	0.836	0.689
AIC	232 325.5	471 395.1
BIC	232 363.3	471 435.7
Log.Lik.	-13 820.998	-33 699.484
F	24 327.769	18 593.859
RMSE	0.64	0.93

* p < 0.1, ** p < 0.05, *** p < 0.01

```
form_ind <- as.formula("log(Y2) ~ industry + log_L:industry + log_K:industry + log_i:industry")
ols_inv_ind_col <- lm(form_ind, data = data_col)
ols_inv_ind_chi <- lm(form_ind, data = data_chi)

modelsummary(
  list(Colombia_Ind = ols_inv_ind_col, Chile_Ind = ols_inv_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

2.3

```
# First stage: estimate log_y2 = log_l + phi(log_k) s.t. phi = alpha_0 +
# alpha_k*log_k + a(i*t*k)^2 + b(i*t*k) + c

data_col <- data_col %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
```

	Colombia_Ind	Chile_Ind
(Intercept)	2.545*** (0.048)	3.106*** (0.084)
industryfood	−0.161*** (0.061)	−1.015*** (0.091)
industrymetal	−0.504*** (0.072)	0.101 (0.109)
industrytext	−0.267*** (0.074)	0.253** (0.108)
industrywood	−0.027 (0.114)	−0.482*** (0.113)
industryapparel × log_L	0.925*** (0.015)	1.006*** (0.027)
industryfood × log_L	0.767*** (0.011)	0.843*** (0.012)
industrymetal × log_L	0.985*** (0.017)	0.943*** (0.022)
industrytext × log_L	0.834*** (0.017)	0.846*** (0.020)
industrywood × log_L	0.954*** (0.034)	0.977*** (0.025)
industryapparel × log_K	0.119*** (0.012)	0.146*** (0.017)
industryfood × log_K	0.267*** (0.010)	0.339*** (0.006)
industrymetal × log_K	0.196*** (0.014)	0.218*** (0.012)
industrytext × log_K	0.230*** (0.014)	0.202*** (0.013)
industrywood × log_K	0.139*** (0.028)	0.176*** (0.014)
industryapparel × log_i	0.015** (0.007)	0.016*** (0.004)
industryfood × log_i	0.042*** (0.007)	0.015*** (0.003)
industrymetal × log_i	0.017* (0.009)	0.022*** (0.005)
industrytext × log_i	0.045*** (0.009)	0.027*** (0.004)
industrywood × log_i	−0.001 (0.016)	0.020*** (0.005)
Num.Obs.	14 315	25 125
R2	0.843	0.715
R2 Adj.	0.843	0.715
AIC	231 710.0	469 242.0
BIC	231 869.0	469 412.8
Log.Lik.	−13 497.278	−32 606.965
F	4049.263	3320.642
RMSE	0.62	0.89

* p < 0.1, ** p < 0.05, *** p < 0.01


```

    log_k_next = lead(log_K),
    log_y2_next = lead(log_Y2)
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    log_k_next = lead(log_K),
    log_y2_next = lead(log_Y2)
  ) %>%
  ungroup()

# Define a function to perform the first stage regression and get phi_hat
first_stage <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      inv_sq = log_i^2,
      k_inv = log_K * log_i,
      k_inv_sq = k_inv^2,
      rhs = cbind(log_L, log_K, k_sq, log_i, inv_sq, k_inv, k_inv_sq)
    )
  fs_model <- lm(log_Y2 ~ log_L + rhs, data = data)
  data$bl_hat <- coef(fs_model)["log_L"]
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}
data_col <- first_stage(data_col)
data_chi <- first_stage(data_chi)

# Second stage: estimate  $y_{2,t+1} - \alpha_l l_{t+1} = \alpha_0 + \alpha_k k_{t+1}$ 
# +  $g(\phi_{hat,t} - \alpha_k k_t - \alpha_0) + \text{error}$ 

data_col <- data_col %>%
  mutate(
    lhs = log_y2_next - bl_hat * lead(log_L)
  )
data_chi <- data_chi %>%
  mutate(

```

	Colombia	Chile
beta0	0.839*** (0.069)	0.755*** (0.020)
betak	0.218*** (0.042)	0.440*** (0.037)
betag	0.510*** (0.006)	0.522*** (0.006)
Num.Obs.	14 273	25 099
AIC	31 714.9	69 831.0
BIC	31 745.2	69 863.5
Log.Lik.	-15 853.475	-34 911.508
isConv	TRUE	TRUE
finTol	5.87543307232137e-08	2.51406477231752e-09

* p < 0.1, ** p < 0.05, *** p < 0.01

```

    lhs = log_y2_next - bl_hat * lead(log_L)
  )

form_op <- as.formula("lhs ~ beta0 * betak*log_k_next + betag*I(phi_hat - betak*log_K)")

op_col <- nls(form_op, data = data_col,
  start = list(beta0 = 0, betak = 0.3, betag = 0.3))
op_chi <- nls(form_op, data = data_chi,
  start = list(beta0 = 0, betak = 0.3, betag = 0.3))

modelsummary(
  list(Colombia = op_col, Chile = op_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

```

# Estimate varying parameters across industries using OP for each country
# First stage already done above
data_col <- data_col %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    lhs = log_y2_next - bl_hat * lead(log_L)
  )
data_chi <- data_chi %>%

```

```

mutate(
  industry_factor = as.factor(industry),
  ind = as.integer(industry_factor),
  lhs = log_y2_next - bl_hat * lead(log_L)
)

n_ind_col <- nlevels(data_col$industry_factor)
n_ind_chi <- nlevels(data_chi$industry_factor)

form_op_ind <- as.formula("lhs ~ beta0[ind] + betak[ind]*log_k_next + betag[ind]*I(phi_hat -
op_ind_col <- nls(form_op_ind, data = data_col,
  start = list(beta0 = rep(0, n_ind_col), betak = rep(0.3, n_ind_col), betag = rep(0.3, n_in
op_ind_chi <- nls(form_op_ind, data = data_chi,
  start = list(beta0 = rep(0, n_ind_chi), betak = rep(0.3, n_ind_chi), betag = rep(0.3, n_in

modelsummary(
  list(Colombia_Ind = op_ind_col, Chile_Ind = op_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

2.4

```

first_stage_lp <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      inv_sq = log_i^2,
      k_inv = log_K * log_i,
      k_inv_sq = k_inv^2,
      log_M = log(M),
      log_M_next = lead(log_M),
      m_sq = log_M^2,
      k_m = log_K * log_M,
      k_m_sq = k_m^2,
      m_inv = log_M * log_i,
      m_inv_sq = m_inv^2,
      rhs = cbind(log_L, log_K, k_sq, log_M, m_sq, log_i, inv_sq,

```

	Colombia_Ind	Chile_Ind
beta01	1.947*** (0.073)	2.053*** (0.118)
beta02	1.944*** (0.046)	1.356*** (0.057)
beta03	1.227*** (0.072)	2.480*** (0.104)
beta04	1.807*** (0.072)	2.936*** (0.089)
beta05	1.746*** (0.147)	1.759*** (0.115)
betak1	0.093*** (0.015)	0.064* (0.035)
betak2	0.279*** (0.008)	0.357*** (0.007)
betak3	0.165*** (0.019)	0.218*** (0.019)
betak4	0.262*** (0.013)	0.235*** (0.016)
betak5	0.058* (0.035)	0.152*** (0.025)
betag1	0.214*** (0.018)	0.337*** (0.032)
betag2	0.092*** (0.014)	0.154*** (0.015)
betag3	0.309*** (0.022)	0.228*** (0.027)
betag4	0.122*** (0.021)	0.080*** (0.024)
betag5	0.288*** (0.039)	0.263*** (0.031)
Num.Obs.	14 273	25 099
AIC	28 118.9	65 007.5
BIC	28 239.9	65 137.6
Log.Lik.	-14 043.439	-32 487.742
isConv	TRUE	TRUE
finTol	6.64993877445889e-07	4.33991608884684e-06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

```

      k_inv, k_inv_sq, k_m, k_m_sq, m_inv, m_inv_sq)
    )
    fs_model <- lm(log_Y2 ~ log_L + rhs, data = data)
    data$bl_hat <- coef(fs_model)["log_L"]
    data$phi_hat <- predict(fs_model, newdata = data)
    return(data)
  }

data_col_lp <- first_stage_lp(data_col)
data_chi_lp <- first_stage_lp(data_chi)

data_col_lp <- data_col_lp %>%
  mutate(
    log_y2_next = lead(log_Y2),
    log_k_next = lead(log_K),
    log_M_next = lead(log_M),
    lhs = log_y2_next - bl_hat * lead(log_L)
  )
data_chi_lp <- data_chi_lp %>%
  mutate(
    log_y2_next = lead(log_Y2),
    log_k_next = lead(log_K),
    log_M_next = lead(log_M),
    lhs = log_y2_next - bl_hat * lead(log_L)
  )

# Second stage for LP
form_lp <- as.formula("lhs ~ beta0 + betak*log_k_next + betam*log_M_next + betag*I(phi_hat -
lp_col <- nls(form_lp, data = data_col_lp,
  start = list(beta0 = 0, betak = 0.3, betam = 0.3, betag = 0.3))
lp_chi <- nls(form_lp, data = data_chi_lp,
  start = list(beta0 = 0, betak = 0.3, betam = 0.3, betag = 0.3))

modelsummary(
  list(Colombia = lp_col, Chile = lp_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

# Estimate varying parameters across industries using LP for each country
data_col_lp <- data_col_lp %>%
  mutate(

```

	Colombia	Chile
beta0	1.131*** (0.023)	0.673*** (0.033)
betak	0.083*** (0.005)	0.186*** (0.006)
betam	0.350*** (0.005)	0.396*** (0.008)
betag	0.112*** (0.010)	0.203*** (0.018)
Num.Obs.	14 315	25 125
AIC	23 313.2	63 675.1
BIC	23 351.0	63 715.7
Log.Lik.	-11 651.584	-31 832.539
isConv	TRUE	TRUE
finTol	1.55816414427135e-06	7.12031920263528e-06

* p < 0.1, ** p < 0.05, *** p < 0.01

```

    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    log_M_next = lead(log_M)
  )
data_chi_lp <- data_chi_lp %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    log_M_next = lead(log_M)
  )

n_ind_col <- nlevels(data_col_lp$industry_factor)
n_ind_chi <- nlevels(data_chi_lp$industry_factor)

form_lp_ind <- as.formula("lhs ~ beta0[ind] + betak[ind]*log_k_next + betam[ind]*log_M_next + betag[ind]*log_g_next")
lp_ind_col <- nls(form_lp_ind, data = data_col_lp,
  start = list(beta0 = rep(0, n_ind_col), betak = rep(0.3, n_ind_col), betam = rep(0.3, n_ind_col), betag = rep(0.3, n_ind_col))
lp_ind_chi <- nls(form_lp_ind, data = data_chi_lp,
  start = list(beta0 = rep(0, n_ind_chi), betak = rep(0.3, n_ind_chi), betam = rep(0.3, n_ind_chi), betag = rep(0.3, n_ind_chi))

modelsummary(
  list(Colombia_Ind = lp_ind_col, Chile_Ind = lp_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),

```

```

    output = "kableExtra"
  )

```

2.5

```

acf_fs <- function(data) {
  data <- data %>%
    arrange(id, year) %>%
    group_by(id) %>%
    filter(
      !is.na(log_Y2),
      !is.na(log_L),
      !is.na(log_K),
      !is.na(M)
    ) %>%
    mutate(
      lag_log_L = lag(log_L, 1),
      lag_log_K = lag(log_K, 1),
      t2_lag_log_L = lag(log_L, 2),
      t2_lag_log_K = lag(log_K, 2)
    ) %>%
    filter(
      !is.na(lag_log_L),
      !is.na(lag_log_K),
      !is.na(t2_lag_log_L),
      !is.na(t2_lag_log_K)
    ) %>%
    ungroup() %>%
    mutate(
      k_sq = log_K^2,
      l_sq = log_L^2,
      l_k = log_L * log_K,
      l_k_sq = l_k^2,
      log_M = log(M),
      m_sq = log_M^2,
      l_m = log_L * log_M,
      l_m_sq = l_m^2,
      k_m = log_K * log_M,
      k_m_sq = k_m^2,
      l_k_m = log_L * log_K * log_M,

```

	Colombia_Ind	Chile_Ind
beta01	1.403*** (0.056)	1.153*** (0.098)
beta02	0.976*** (0.036)	0.014 (0.045)
beta03	1.040*** (0.054)	1.456*** (0.084)
beta04	1.086*** (0.056)	1.536*** (0.084)
beta05	1.309*** (0.114)	0.648*** (0.094)
betak1	0.029** (0.012)	0.009 (0.039)
betak2	0.074*** (0.007)	0.155*** (0.008)
betak3	0.067*** (0.011)	0.113*** (0.020)
betak4	0.118*** (0.009)	0.136*** (0.019)
betak5	0.047* (0.025)	0.075*** (0.023)
betam1	0.267*** (0.013)	0.221*** (0.032)
betam2	0.412*** (0.007)	0.495*** (0.010)
betam3	0.434*** (0.013)	0.372*** (0.022)
betam4	0.451*** (0.011)	0.275*** (0.025)
betam5	0.237*** (0.033)	0.441*** (0.022)
betag1	0.216*** (0.021)	0.518*** (0.037)
betag2	0.010 (0.016)	0.177*** (0.025)
betag3	0.037 (0.029)	0.308*** (0.043)
betag4	-0.143*** (0.027)	0.308*** (0.037)
betag5	0.275*** (0.048)	0.244*** (0.052)
Num.Obs.	14 315	25 125
AIC	22 553.0	60 324.9
BIC	22 712.0	60 495.7
Log.Lik.	-11 255.502 ₂₄	-30 141.452
isConv	TRUE	TRUE
finTol	1.58192225482312e-06	7.95736960046656e-06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$


```

    l_k_m_sq = l_k_m^2

  )

fs_model <- lm(
  log_Y2 ~ log_L + log_K + log_M + k_sq + l_sq + l_k + l_k_sq +
    m_sq + l_m + l_m_sq + k_m + k_m_sq + l_k_m + l_k_m_sq,
  data = data
)

data$phi_hat <- as.numeric(predict(fs_model, newdata = data))
data
}

data_col_acf <- acf_fs(data_col)
data_chi_acf <- acf_fs(data_chi)

acf_nls <- function(df) {
  df_nls <- df %>%
    filter(
      !is.na(log_Y2),
      !is.na(log_L),
      !is.na(log_K),
      !is.na(phi_hat),
      !is.na(lag_log_K),
      !is.na(lag_log_L),
      !is.na(t2_lag_log_L)
    ) %>%
    mutate(
      lag_phi_hat = lag(phi_hat, 1)
    )

  start_vals <- list(
    beta0 = 0,
    betal = 0.3,
    betak = 0.3,
    betag = 0.7
  )

  nls(
    log_Y2 ~ beta0 + betal * log_L + betak * log_K + betag * (lag_phi_hat - (betak * lag_
    data = df_nls,

```

	Colombia	Chile
beta0	1.473*** (0.029)	1.525*** (0.030)
betal	0.814*** (0.009)	0.885*** (0.010)
betak	0.317*** (0.006)	0.344*** (0.005)
betag	0.244*** (0.007)	0.249*** (0.007)
Num.Obs.	12 710	28 911
AIC	25 120.2	77 711.9
BIC	25 157.4	77 753.3
Log.Lik.	-12 555.076	-38 850.964
isConv	TRUE	TRUE
finTol	1.6502288794746e-06	8.94887350168634e-06

* p < 0.1, ** p < 0.05, *** p < 0.01

```

      start = start_vals
    )
  }

acf_col <- acf_nls(data_col_acf)
acf_chi <- acf_nls(data_chi_acf)

modelsummary(
  list(Colombia = acf_col, Chile = acf_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

```

# Estimate varying parameters across industries using ACF for each country
data_col_acf <- data_col_acf %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    lag_phi_hat = lag(phi_hat, 1)
  )
data_chi_acf <- data_chi_acf %>%
  mutate(
    industry_factor = as.factor(industry),

```

```

    ind = as.integer(industry_factor),
    lag_phi_hat = lag(phi_hat, 1)
  )
n_ind_col <- nlevels(data_col_acf$industry_factor)
n_ind_chi <- nlevels(data_chi_acf$industry_factor)

form_acf_ind <- as.formula("log_Y2 ~ beta0[ind] + betal[ind] * log_L + betak[ind] * log_K + l")
acf_ind_col <- nls(form_acf_ind, data = data_col_acf,
  start = list(beta0 = rep(0, n_ind_col), betal = rep(0.3, n_ind_col), betak = rep(0.3, n_ind_col))
)
acf_ind_chi <- nls(form_acf_ind, data = data_chi_acf,
  start = list(beta0 = rep(0, n_ind_chi), betal = rep(0.3, n_ind_chi), betak = rep(0.3, n_ind_chi))
)
modelsummary(
  list(Colombia_Ind = acf_ind_col, Chile_Ind = acf_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)

```

Question 3

3.1

```

# Using OP estimates from 2.3 to estimate omega = exp(phi_it - alpha_0 - alpha_k*log(K_it))

alpha_0_col <- coef(op_col)[["beta0"]]
alpha_k_col <- coef(op_col)[["betak"]]
data_col <- data_col %>%
  mutate(
    omega_it = exp(phi_hat - alpha_0_col - alpha_k_col * log_K)
  )
alpha_0_chi <- coef(op_chi)[["beta0"]]
alpha_k_chi <- coef(op_chi)[["betak"]]
data_chi <- data_chi %>%
  mutate(
    omega_it = exp(phi_hat - alpha_0_chi - alpha_k_chi * log_K)
  )
data_col <- data_col %>%
  mutate(
    mean_omega_col = mean(omega_it, na.rm = TRUE),
    sd_omega_col = sd(omega_it, na.rm = TRUE)
  )

```

	Colombia_Ind	Chile_Ind
beta01	2.224*** (0.069)	2.268*** (0.098)
beta02	1.200*** (0.047)	1.068*** (0.038)
beta03	1.436*** (0.070)	2.395*** (0.079)
beta04	1.702*** (0.071)	2.232*** (0.083)
beta05	2.046*** (0.145)	1.948*** (0.088)
beta11	0.921*** (0.018)	1.086*** (0.028)
beta12	0.677*** (0.017)	0.893*** (0.015)
beta13	0.970*** (0.022)	1.013*** (0.027)
beta14	0.812*** (0.021)	0.858*** (0.024)
beta15	1.057*** (0.042)	1.042*** (0.027)
betak1	0.140*** (0.012)	0.149*** (0.017)
betak2	0.402*** (0.011)	0.383*** (0.007)
betak3	0.263*** (0.015)	0.237*** (0.014)
betak4	0.321*** (0.014)	0.278*** (0.014)
betak5	0.109*** (0.024)	0.181*** (0.015)
betag1	0.107*** (0.017)	0.179*** (0.024)
betag2	0.352*** (0.012)	0.289*** (0.010)
betag3	0.178*** (0.018)	0.210*** (0.019)
betag4	0.131*** (0.017)	0.219*** (0.020)
betag5	0.098*** (0.038)	0.140*** (0.019)
Num.Obs.	12 710	28 911
AIC	24 551.5	74 842.5
BIC	24 708.0	75 016.2
Log.Lik.	-12 254.749 ₂₈	-37 400.267
isConv	TRUE	TRUE
finTol	1.93813455876023e-06	4.32316234385992e-06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

```
data_chi <- data_chi %>%
  mutate(
    mean_omega_chi = mean(omega_it, na.rm = TRUE),
    sd_omega_chi = sd(omega_it, na.rm = TRUE)
  )

print(paste("Colombia - Mean Omega:", data_col$mean_omega_col[1], "SD Omega:", data_col$sd_omega_col[1]))
```

```
[1] "Colombia - Mean Omega: 294.293581267125 SD Omega: 702.900302599767"
```

```
print(paste("Chile - Mean Omega:", data_chi$mean_omega_chi[1], "SD Omega:", data_chi$sd_omega_chi[1]))
```

```
[1] "Chile - Mean Omega: 74.8760195478157 SD Omega: 82.6386074307829"
```

3.2

```
## Colombia: extract industry-specific OP parameters
coef_col_ind <- coef(op_ind_col)

beta0_col_ind <- coef_col_ind[grep("^beta0", names(coef_col_ind))]
betak_col_ind <- coef_col_ind[grep("^betak", names(coef_col_ind))]
# betag_col_ind <- coef_col_ind[grep("^betag", names(coef_col_ind))] # not needed for omega

# Make sure 'ind' matches the order of industry_factor levels used in nls
# ind = as.integer(industry_factor) is already set upstream

data_col <- data_col %>%
  mutate(
    alpha0_ind = beta0_col_ind[ind],
    alphaK_ind = betak_col_ind[ind],
    omega_it_het = exp(phi_hat - alpha0_ind * 1 - alphaK_ind * log_K)
  )

## Chile
coef_chi_ind <- coef(op_ind_chi)

beta0_chi_ind <- coef_chi_ind[grep("^beta0", names(coef_chi_ind))]
betak_chi_ind <- coef_chi_ind[grep("^betak", names(coef_chi_ind))]
```

```

data_chi <- data_chi %>%
  mutate(
    alpha0_ind = beta0_chi_ind[ind],
    alphaK_ind = betak_chi_ind[ind],
    omega_it_het = exp(phi_hat - alpha0_ind * 1 - alphaK_ind * log_K)
  )

omega_industry_col <- data_col %>%
  group_by(industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega    = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )

omega_industry_chi <- data_chi %>%
  group_by(industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega    = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )

print("Colombia - Mean and SD Omega by Industry:")

```

```
[1] "Colombia - Mean and SD Omega by Industry:"
```

```
print(omega_industry_col)
```

```

# A tibble: 5 x 3
  industry mean_omega sd_omega
  <chr>      <dbl>    <dbl>
1 apparel    163.     376.
2 food       61.7     115.
3 metal     238.     400.
4 text      124.     279.
5 wood      294.    1310.

```

```
print("Chile - Mean and SD Omega by Industry:")
```

```
[1] "Chile - Mean and SD Omega by Industry:"
```

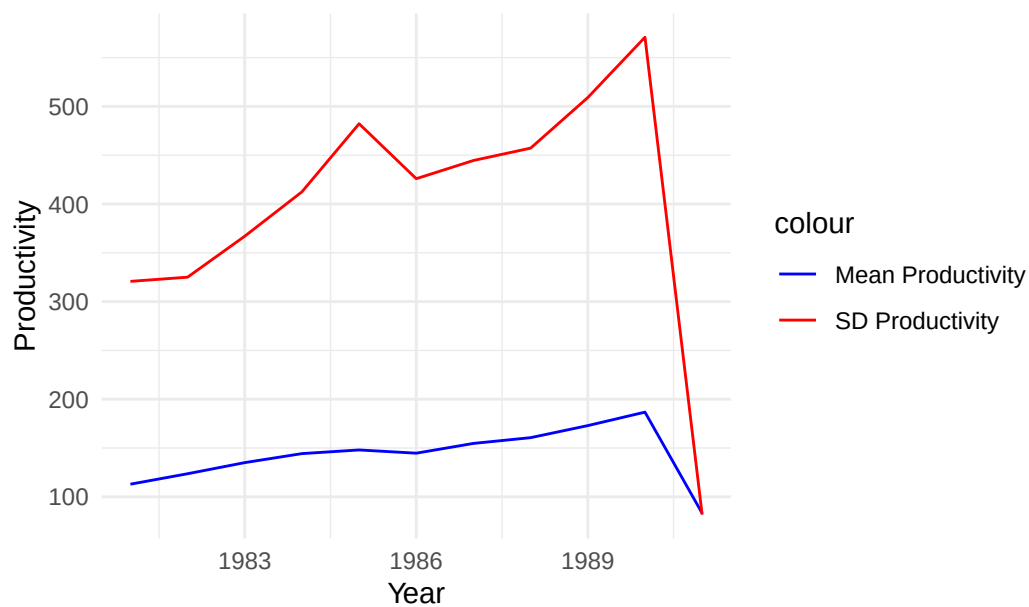
```
print(omega_industry_chi)
```

```
# A tibble: 5 x 3
  industry mean_omega sd_omega
  <chr>      <dbl>    <dbl>
1 apparel    460.    1849.
2 food       90.9     124.
3 metal      88.7     119.
4 text       62.7     134.
5 wood      357.     619.
```

3.3

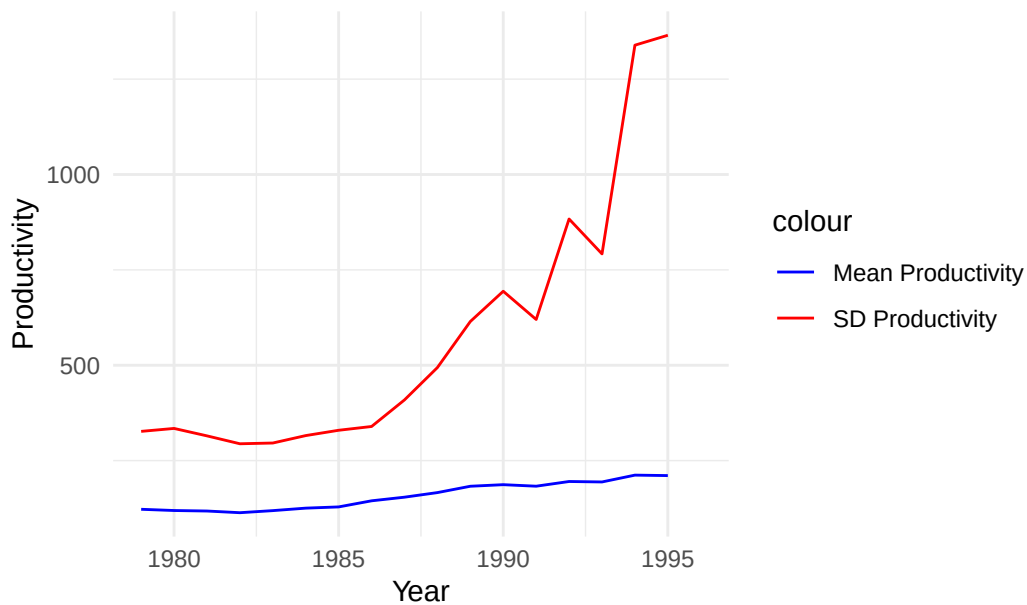
```
# Calculate yearly mean and sd of omega_it_het for each country
prod_trends_col <- data_col %>%
  group_by(year) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega    = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
prod_trends_chi <- data_chi %>%
  group_by(year) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega    = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
# Plot trends over time for each country
ggplot(prod_trends_col, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = "Mean Productivity")) +
  geom_line(aes(y = sd_omega, color = "SD Productivity")) +
  labs(title = "Trends in Productivity over Time (Colombia)",
       x = "Year",
       y = "Productivity") +
  scale_color_manual(values = c("Mean Productivity" = "blue", "SD Productivity" = "red")) +
  theme_minimal()
```

Trends in Productivity over Time (Colombia)



```
ggsave("3.3_hom_col.pdf", width = 10, height = 6)
ggplot(prod_trends_chi, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = "Mean Productivity")) +
  geom_line(aes(y = sd_omega, color = "SD Productivity")) +
  labs(title = "Trends in Productivity over Time (Chile)",
        x = "Year",
        y = "Productivity") +
  scale_color_manual(values = c("Mean Productivity" = "blue", "SD Productivity" = "red")) +
  theme_minimal()
```


Trends in Productivity over Time (Chile)

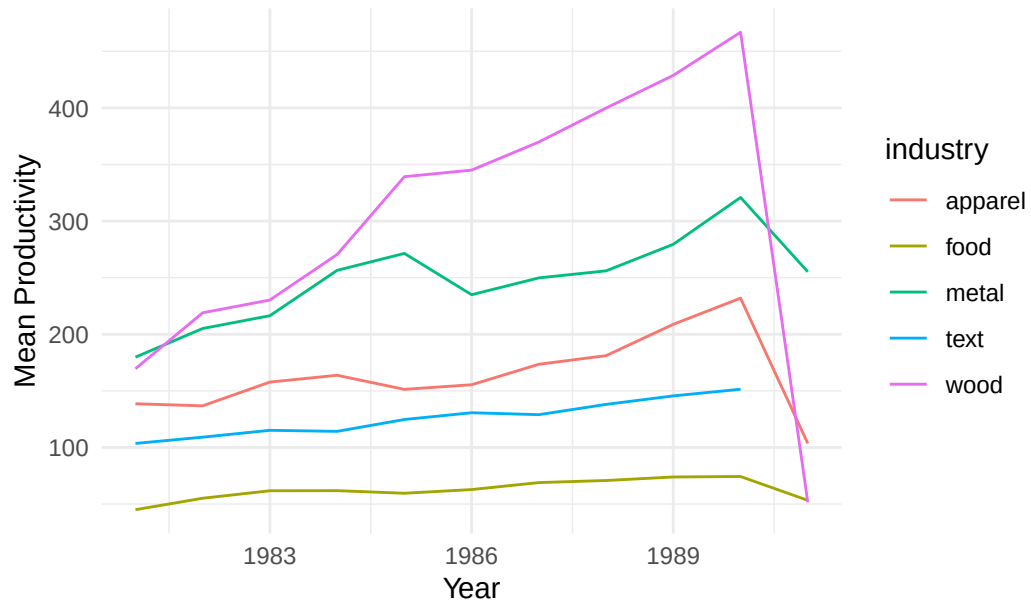


```
ggsave("3.3_hom_chi.pdf", width = 10, height = 6)

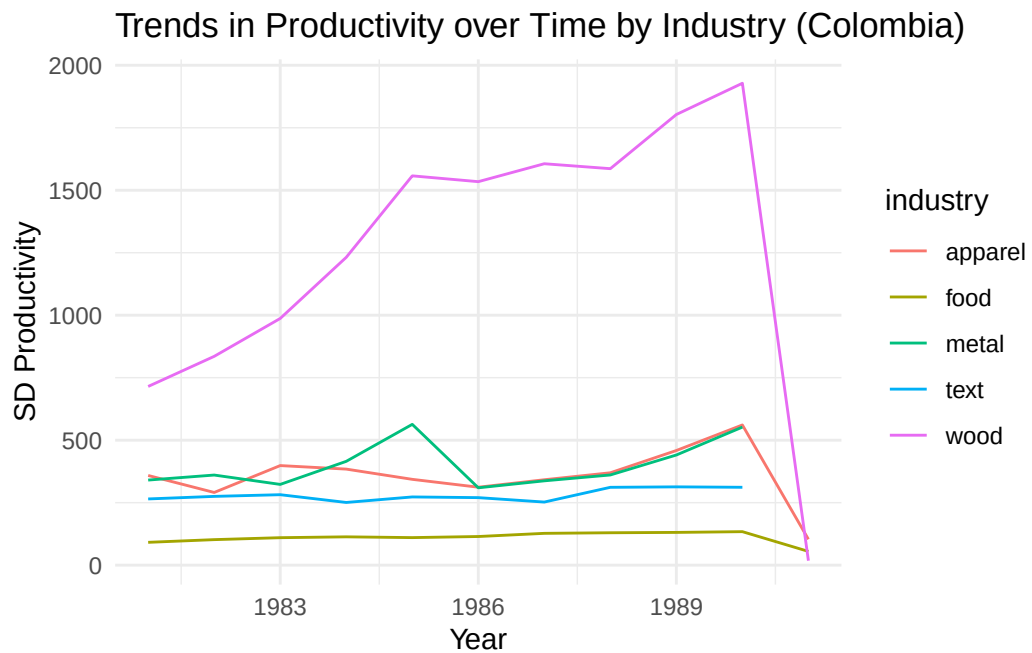
# Now by industry for each country
prod_trends_ind_col <- data_col %>%
  group_by(year, industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega    = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
prod_trends_ind_chi <- data_chi %>%
  group_by(year, industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega    = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )

# Plot trends over time by industry for each country
ggplot(prod_trends_ind_col, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Colombia)",
       x = "Year",
       y = "Mean Productivity") +
  theme_minimal()
```

Trends in Productivity over Time by Industry (Colombia)

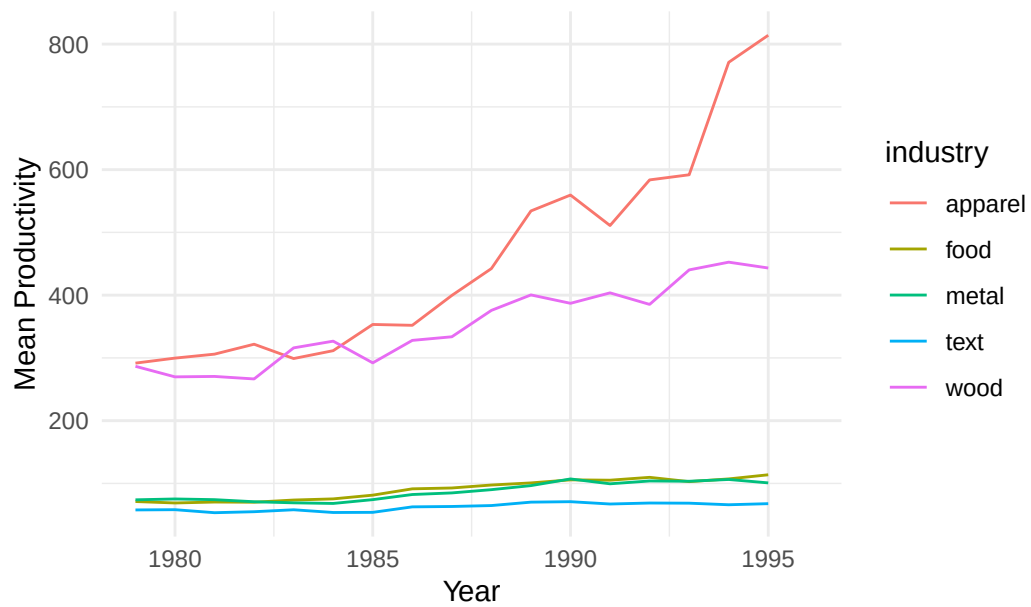


```
ggsave("3.3_ind_col_mean.pdf", width = 10, height = 6)
ggplot(prod_trends_ind_col, aes(x = year)) +
  geom_line(aes(y = sd_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Colombia)",
        x = "Year",
        y = "SD Productivity") +
  theme_minimal()
```



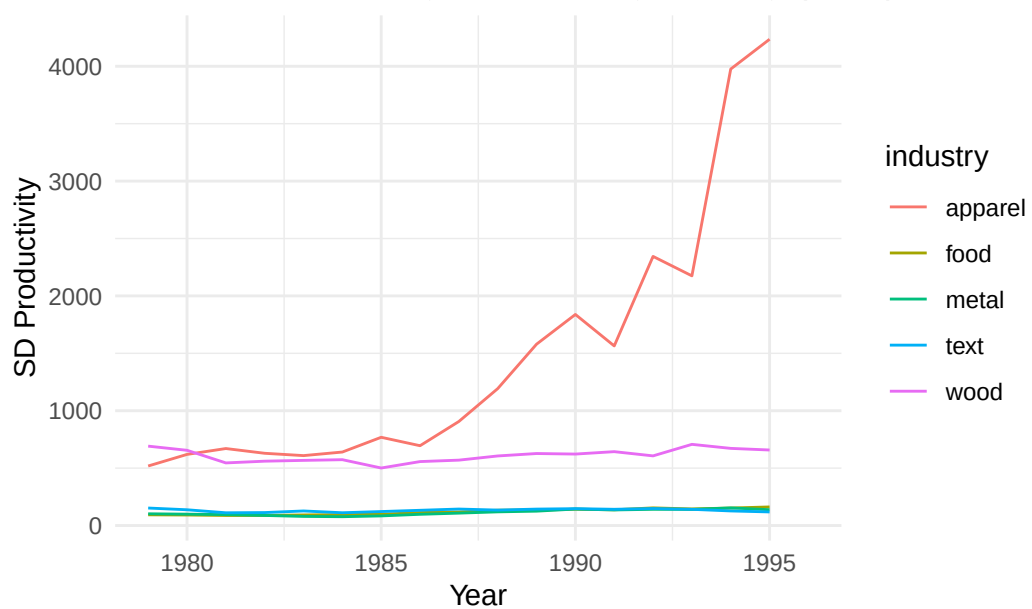
```
ggsave("3.3_ind_col_sd.pdf", width = 10, height = 6)
ggplot(prod_trends_ind_chi, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Chile)",
        x = "Year",
        y = "Mean Productivity") +
  theme_minimal()
```

Trends in Productivity over Time by Industry (Chile)



```
ggsave("3.3_ind_chi_mean.pdf", width = 10, height = 6)
ggplot(prod_trends_ind_chi, aes(x = year)) +
  geom_line(aes(y = sd_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Chile)",
        x = "Year",
        y = "SD Productivity") +
  theme_minimal()
```

Trends in Productivity over Time by Industry (Chile)



```
ggsave("3.3_ind_chi_sd.pdf", width = 10, height = 6)
```